A Rate-Distortion Function for Model Merging

Sankalp Pathak
pathaksankalp04@gmail.com

Sanjay Garg
gargsv@gmail.com

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Abstract

LoRA fine-tunes are routinely merged into a single deployable model, but no principled limit governs how well this can be done at a fixed storage budget. We pose model merging as multi-source lossy source coding under worst-task distortion and establish the first rate-distortion theorem for the problem. Our main tool is an H_t-weighted max-$\geq$-avg identity that reduces worst-task distortion to average-distortion quantization of an H_t-weighted centroid plus an irreducible floor. Under a Stiefel-random hard distribution with arbitrary task-dependent curvatures, the floor admits a closed form $B^2(1 - d_{\text{eff}}/(Tr))$ in terms of the effective dimension $d_{\text{eff}} := \mathbb{E}[\text{rank}(\sum_t P_{V_t})]$, and the compression term scales as $\Theta(2^{-2R/d_{\text{eff}}})$. We prove a matching achievability bound via explicit Gaussian-QR random orthogonal mixing followed by uniform scalar quantization, pinning down the rate-distortion function $\mathcal{D}^\star = \Theta(B^2 \cdot 2^{-2R/(Tr)})$ up to a universal constant $C = O(T)$ in the generic floor-zero regime. Monte Carlo experiments validate the predicted slope to within ± 0.01 across $T \in \{2, 3, 4, 8\}$ and a range of anisotropic spectra. The bound gives a calibration curve against which existing and future merging algorithms can be graded, and explains a recent large-scale empirical finding that subspace-alignment methods fail at LLM scale when the floor term is bounded away from zero.

1 Introduction

Model merging is everywhere, but nobody knows the limit. Low-Rank Adaptation [6] has made it routine to maintain dozens of task-specific fine-tunes per base model, and it has made *merging* those fine-tunes into a single deployable model the dominant practical question. A now-standard workflow: fine-tune T LoRA adapters on T downstream tasks, combine them post-hoc, serve one model. Task Arithmetic [8], TIES [19], DARE [22], KnOTS [17], DO-Merging [25], Core Space [15], Task-Vector Quantization [10], and 1-bit-Merging [11] all compete on this problem. Each paper proposes an encoder, tunes some hyper-parameters, and reports benchmark numbers. None answers the prior question: given a storage budget of R bits per merged model, what is the *best possible* merging algorithm, and how far are existing methods from it?

The rate-distortion question. We formalize this as multi-source lossy source coding. Fix T tasks, ambient dimension d, and LoRA rank $r \leq d$. Each task is characterized by a pair (τ_t, H_t): a task vector $\tau_t \in \mathbb{R}^d$ (the LoRA update) and a positive-semidefinite H_t of rank r (the task's Hessian at τ_t, or a Fisher surrogate) with $\tau_t^\top H_t \tau_t \leq B^2$. Given such a tuple, an encoder maps $(\tau_1, \ldots, \tau_T)$ into R bits; a decoder reconstructs a single merged model $w^\star \in \mathbb{R}^d$. Per-task distortion is $D_t(w) := (w - \tau_t)^\top H_t (w - \tau_t)$, the second-order loss inflation from using w in place of τ_t. The figure of merit is the *worst-task* distortion $\max_{t \leq T} D_t(w^\star)$ — a merged model is only as good as its worst task. The rate-distortion function $\mathcal{D}^\star(T, d, r, B, R)$ is the infimum over encoders and decoders of expected max-distortion at rate R.

Two features separate this problem from classical lossy source coding. First, the encoder observes T sources rather than one; multi-description coding [4] addresses related questions but under *average* distortion, which Lemma 1 (below) shows is strictly weaker. Second, the low-rank structure of H_t means distortions are measured in task-specific rank-r subspaces; the effective dimension is the combinatorial quantity $d_{\text{eff}} := \mathbb{E}[\text{rank}(\sum_t P_{V_t})]$, which satisfies $d_{\text{eff}} \leq \min(Tr, d)$, with the bound attained almost surely when the V_t's are drawn iid from the Stiefel manifold.

Contributions. We give the first rate-distortion theorem for model merging, matching lower and upper bounds up to universal constants in the generic regime.

1. **H_t-weighted max $\geq$ avg identity (Lemma 1).** For every admissible (τ, H) and every $w \in \mathbb{R}^d$, $\max_t D_t(w) \geq (w - \bar{\tau}_H)^\top \bar{H} (w - \bar{\tau}_H) + (1/T) \sum_t (\tau_t - \bar{\tau}_H)^\top H_t (\tau_t - \bar{\tau}_H)$, where $\bar{\tau}_H = \bar{H}^+ \cdot T^{-1} \sum_t H_t \tau_t$ is the H_t-weighted centroid and $\bar{H} = T^{-1} \sum_t H_t$. This identity reduces max-distortion merging to average-distortion quantization of $\bar{\tau}_H$ plus an irreducible floor. It is the technical bridge from Shannon's single-source rate-distortion machinery to the multi-source max-distortion setting.

2. **Closed-form floor (Lemma 2).** Under a Stiefel-random hard distribution $P^\star$ on (τ, H), the floor term reduces to $B^2(1 - d_{\text{eff}}/(Tr))$ exactly. The interpolation between $T = 1$ (floor 0, all bits go to compression) and V_t mutually orthogonal (floor 0 again, reconstructive task arithmetic is lossless at infinite rate) is explicit.

3. **Rate-distortion lower bound (Theorem 3).** Under $P^\star$ with arbitrary positive-definite spectra $(D_t)_{t=1}^T$,

$$\mathcal{D}^\star(T, d, r, B, R) \;\geq\; B^2\big(1 - \tfrac{d_{\text{eff}}}{Tr}\big) + c_{\text{TQ}} \cdot \tfrac{B^2 d_{\text{eff}}}{Tr} \cdot 2^{-2R/d_{\text{eff}}}.$$

   The proof composes Yao's minimax principle [23], Lemma 1, a reduction to the range$(\bar{H})$ subspace, and an $O(d)$-equivariance argument that works regardless of task-dependent spectra.

4. **Matching achievability (Theorem 4).** In the floor-zero Stiefel regime ($Tr \leq d$, V_t drawn iid from the Stiefel manifold), an explicit algorithm —

randomized orthogonal mixing of the $\bar{H}^{1/2}$-scaled centroid followed by uniform scalar quantization — achieves $\mathbb{E}[\max_t D_t(w^\star)] \leq C \cdot B^2 \cdot 2^{-2R/(Tr)}$ for a universal constant $C = O(T)$. Combined with Theorem 3, $\mathcal{D}^\star = \Theta\big(B^2 \cdot 2^{-2R/(Tr)}\big)$ in this regime. The construction reduces to TurboQuant [23] at $T = 1$.

5. **Numerical validation.** Monte Carlo experiments across $T \in \{2, 3, 4, 8\}$, $d \in \{128, 512, 2048\}$, $r \in \{4, 8, 16\}$, with both isotropic and heavily anisotropic task-dependent spectra, confirm the predicted slope $-2R/d_{\text{eff}}$ on $\log_2 \mathbb{E}[\text{excess}]$ to within bootstrap 95% CI ± 0.01 across 1000 trials per configuration. The achievability constant C is empirically $\approx$ 11–13, matching the analytical bound $Tc^2/3$.

What this means for practice. The bound gives practitioners a calibration curve. Any merging algorithm — existing or future — can be graded by its gap to $\mathcal{D}^\star$. Two qualitative implications are already visible from the theorem statement. (i) The floor $B^2(1 - d_{\text{eff}}/(Tr))$ is irreducible: when V_t's overlap substantially ($d_{\text{eff}} \ll Tr$), no amount of rate closes it. This recovers and explains the recent large-scale empirical finding of [5] that subspace-based merging methods fail at LLM scale — random LoRA checkpoints have substantial subspace overlap, so the floor is bounded away from zero. (ii) Additional rate helps at the $2^{-2R/d_{\text{eff}}}$ rate, matching what TurboQuant achieves in the single-source case. A practitioner storing a merged model at 4 bits per coordinate operates roughly $4d_{\text{eff}}$-bits-below the infinite-rate floor; doubling to 8 bits gains an $\approx 16\times$ reduction in compression excess.

Scope and limitations. The matching upper bound in Theorem 4 is proved under the generic Stiefel-random hard distribution $P^\star$ with $Tr \leq d$. In the floor-positive regime (shared or heavily overlapping V_t), an explicit linear encoder achieves an excess rate of $2^{-2R/(d_{\text{eff}}+|A|-1)}$ (where $|A|$ is the size of the active set at the Chebyshev center), and we verify this numerically for $T \in \{2, 3, 4\}$; closing the exponent gap to the lower bound's $2^{-2R/d_{\text{eff}}}$ appears to require non-linear / random-coding encoders and is left open (Remark 5). The bound is stated in the MSE / quadratic-loss setting; the extension to cross-entropy LLM loss via local Fisher-quadratic approximation is discussed in §7.

Paper organization. §3 introduces the formal rate-distortion model for merging and the admissible hard distributions. §4 proves the lower bound (Theorem 3) in six steps, with the max-$\geq$-avg identity and closed-form floor as the central tools. §5 gives the matching achievability construction and analyses. §6 reports synthetic experiments on rank-r LoRA-like ensembles validating the theoretical slopes and constants. §7 discusses extensions to cross-entropy loss, the open shared-V exponent gap, and implications for practical merging algorithm design.

2 Related Work

Model merging: constructions without limits. A wave of post-hoc model-merging methods has emerged since Task Arithmetic [8] showed that adding fine-tuning deltas of separately-trained task vectors recovers non-trivial multi-task performance. Fisher-weighted averaging [13], sparsification-based approaches (TIES [19], DARE [22]), and more recent rotation- and subspace-alignment families (KnOTS [17], TSPA [1], DO-Merging [25], Core Space [15], TARA [9], ARM [21]) all seek to reduce destructive interference between task vectors before averaging. A parallel line quantizes task vectors at merge time to shrink the storage footprint: Task-Vector Quantization (TVQ) [10] and 1-bit-Merging [11] achieve competitive accuracy at one to four bits per coordinate. What unifies these methods is that they are *constructions*: each proposes a particular encoder and evaluates it empirically. None establishes a *fundamental limit* on merging quality at a given storage or rate budget. This paper supplies that limit, and — in the generic regime — a matching construction.

Existing theoretical analyses of merging. The closest theoretical work on merging falls into three strands. *Weight disentanglement* (Ortiz-Jimenez et al. [14]) identifies the rank-r Hessian structure of LoRA fine-tunes and relates task arithmetic's empirical success to locally-additive task functionals; this provides the $H_t = P_{V_t}$ (and more generally $H_t = U_t D_t U_t^\top$) formalism we adopt in §3, but does not give rate-distortion bounds. *Task-Vector Bases* [24] posits that fine-tuned weights lie in a thin Gaussian shell — a geometric assumption, not information-theoretic. *On Task Vectors and Gradients* [26] interprets task vectors as single-step gradient updates and provides a second-order Taylor-expansion bound on task arithmetic error. ATM [27] takes the noisy-GD interpretation further. None of these works frame merging as lossy source coding or derive Shannon-style limits. Our work is the first to pose merging as a multi-source rate-distortion problem with worst-task distortion as the figure of merit.

Rate-distortion and rotation-for-quantization ancestry. The information-theoretic toolkit we use is standard. Classical rate-distortion theory [16, 3] gives the single-source setting; multi-description coding [4] addresses multi-receiver lossy coding but under average — not maximum — distortion, which precludes direct application to merging (Lemma 1 shows the max-distortion formulation is strictly stronger and exhibits a different RD function). The direct intellectual ancestor of our achievability construction is TurboQuant [23], which establishes Shannon-style lower bounds on randomized vector quantization via Yao's minimax principle and achieves them up to a universal constant via Haar-rotation + uniform scalar quantization. Our Theorem 4 reduces to TurboQuant Theorem 3 in the $T = 1$ single-task limit. A further line of work uses structured rotations (Hadamard / Walsh-Hadamard) as pre-conditioners for quantization in the single-model setting: QuIP [18], QuaRot [2], SpinQuant [12], and RoLoRA [7]. These motivate our choice of randomized orthogonal mixers in the achievability

algorithm but do not address the multi-source / merging setting.

The November 2025 negative result, and why a lower bound still matters. A recent large-scale empirical study [5] reports that across four LLM families and sixteen benchmarks, only Task Arithmetic reliably produces constructive interference; subspace-alignment methods (KnOTS-style) often underperform or fail. This might seem to undermine the case for theoretical work on merging. It does not. Their result is an *upper bound* on specific algorithms — evidence that those constructions are suboptimal at LLM scale. Our Theorem 3 is a *lower bound* on every algorithm — it says that whatever the best merging procedure is, it cannot beat $B^2 f_{\text{floor}} + c_{\text{TQ}}(B^2 d_{\text{eff}}/(Tr)) \cdot 2^{-2R/d_{\text{eff}}}$. The negative result and our lower bound are compatible and, in fact, mutually reinforcing: the lower bound explains *why* generic subspace alignment is insufficient when the floor term is bounded away from zero (irreducible multi-task tension), and identifies the floor-zero regime in which constructive methods (ours, and potentially refined versions of TVQ or KnOTS) can approach the fundamental limit. Fully characterizing the floor-positive regime remains open (Remark 5).

3 Problem Setup and Preliminaries

Task vectors and per-task geometry. Fix integers $T \geq 1$, $d \geq 2$, $1 \leq r \leq d$, and a radius $B > 0$. Each of the T tasks is specified by a pair $(\tau_t, H_t) \in \mathbb{R}^d \times \mathbb{R}^{d \times d}$ with

$$H_t \succeq 0, \quad \operatorname{rank}(H_t) \leq r, \quad \tau_t \in V_t := \operatorname{range}(H_t), \quad \tau_t^\top H_t \tau_t \leq B^2. \tag{1}$$

Per-task loss is the quadratic $f_t(w) := \frac{1}{2}(w - \tau_t)^\top H_t (w - \tau_t)$, which attains zero at any w with $P_{V_t} w = \tau_t$. We write $\boldsymbol{\tau} := (\tau_t)_{t=1}^T$ and $\boldsymbol{H} := (H_t)_{t=1}^T$ for the tuple data, and call any $(\boldsymbol{\tau}, \boldsymbol{H})$ satisfying (1) *admissible*.

The specialization $H_t = P_{V_t}$ (the orthogonal projector onto V_t) recovers the weight-disentangled LoRA setting of Ortiz-Jimenez et al. [14]; the general $H_t \succeq 0$ case subsumes Fisher-metric surrogates for cross-entropy LLM losses with heterogeneous task-specific curvature.

Worst-task distortion. A merged model $w^\star \in \mathbb{R}^d$ incurs per-task distortion $D_t(w^\star) := (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)$. We take as figure of merit the *worst-task* distortion

$$\Delta(w^\star; \boldsymbol{\tau}, \boldsymbol{H}) \;:=\; \max_{t \in [T]} D_t(w^\star). \tag{2}$$

The choice of max over avg matters. A merged model that performs well on most tasks but catastrophically on one is not a useful deployment target; classical multi-description coding [4] treats average distortion and has a strictly weaker rate-distortion function than the one we establish (see Lemma 1).

Rate-R merging codes. A rate-R merging code is a pair (E, D) with encoder $E : (\mathbb{R}^d)^T \to \{1, \dots, 2^R\}$ and decoder $D : \{1, \dots, 2^R\} \to \mathbb{R}^d$. The encoder observes $\boldsymbol{\tau}$ (the H_t's may be assumed known to the encoder as side information; our lower bound holds even if the encoder is told $\boldsymbol{H}$ for free). The reconstruction is $w^\star := D(E(\boldsymbol{\tau}))$. The *rank-$r$ rate-distortion function* is

$$\mathcal{D}^\star(T, d, r, B, R) \;:=\; \inf_{(E,D)} \sup_{(\boldsymbol{\tau},\boldsymbol{H}) \text{ admissible}} \mathbb{E}\big[\Delta(w^\star; \boldsymbol{\tau}, \boldsymbol{H})\big], \tag{3}$$

where the expectation is over any randomness the encoder and decoder share.

Key derived quantities. We will repeatedly use

$$\bar{H} := \tfrac{1}{T}\sum_{t=1}^{T} H_t, \qquad \bar{\tau}_H := \bar{H}^+ \cdot \tfrac{1}{T}\sum_{t=1}^{T} H_t\tau_t, \qquad d_{\text{eff}} := \operatorname{rank}\Big(\sum_{t=1}^{T} P_{V_t}\Big), \tag{4}$$

where $(\cdot)^+$ denotes the Moore–Penrose pseudoinverse. The *H_t-weighted centroid* $\bar{\tau}_H$ is the unique element of $\operatorname{range}(\bar{H}) = V_1 + \cdots + V_T$ satisfying $\bar{H}\bar{\tau}_H = T^{-1}\sum_t H_t\tau_t$; it reduces to the arithmetic mean $\bar{\tau}$ when $H_t = I_d$. The *effective dimension* $d_{\text{eff}} \le \min(Tr, d)$ is the combinatorial quantity that controls the rate-distortion function; its meaning is the number of independent directions the decoder must reconstruct.

Hard distribution $P^\star$. Our lower bound is established against a specific distribution $P^\star$ on admissible tuples. For each task $t \in [T]$, independently:

1. Sample $U_t \in \text{Stiefel}(d, r)$ uniformly (equivalently, take the Gram–Schmidt orthogonalization of a $d \times r$ matrix of iid Gaussians); set $V_t := \operatorname{range}(U_t)$.
2. Fix a positive-definite diagonal $r \times r$ matrix $D_t \succ 0$ (the task-t spectrum inside V_t); set $H_t := U_t D_t U_t^\top$.
3. Sample $u_t \sim \text{Unif}(S^{r-1})$ and set $v_t := B\, D_t^{-1/2} u_t \in \mathbb{R}^r$, $\tau_t := U_t v_t$.

The scale normalization $\tau_t^\top H_t \tau_t = v_t^\top D_t v_t = B^2\|u_t\|^2 = B^2$ holds by construction. The $D_t^{-1/2}$ stretch makes τ_t uniform on the H_t-weighted unit ellipsoid inside V_t, which is what makes the second-moment identity $\mathbb{E}[H_t\tau_t\tau_t^\top H_t | U_t] = (B^2/r)H_t$ hold *regardless of* D_t. This D_t-freeness is the feature that makes Theorem 3 apply to arbitrary task-dependent spectra. For the projector special case $H_t = P_{V_t}$ (i.e. $D_t = I_r$), $P^\star$ reduces to the Stiefel-random LoRA distribution with $\tau_t \sim \text{Unif}(B \cdot S^{r-1} \cap V_t)$.

Yao's minimax bridge. The worst-case rate-distortion function $\mathcal{D}^\star$ in (3) is at least the Bayesian rate-distortion function under any prior. In particular, taking the prior to be $P^\star$:

$$\mathcal{D}^\star(T, d, r, B, R) \;\ge\; \inf_{(E,D)} \mathbb{E}_{(\boldsymbol{\tau},\boldsymbol{H})\sim P^\star}\big[\Delta(w^\star; \boldsymbol{\tau}, \boldsymbol{H})\big]. \tag{5}$$

This is Yao's minimax principle [20] applied to the randomized-code / worst-case-input game. All subsequent lower bounds are proved against the RHS of (5); the LHS inherits them for free.

4 Lower Bound

This section establishes the rank-r rate-distortion lower bound. The argument rests on three ingredients: a deterministic max-$\geq$-avg identity that converts max-distortion into an average-distortion problem plus an irreducible floor (§4.1); a closed-form evaluation of that floor under the hard distribution $P^\star$ (§4.2); and a rate-distortion lower bound on the remaining average-distortion subproblem via an $O(d)$-equivariance argument and a rotationally-invariant Shannon bound (§4.3).

4.1 The H_t-weighted max $\geq$ avg identity

The Phase 0 identity $T^{-1}\sum_t \|w-\tau_t\|^2 = \|w-\bar{\tau}\|^2 + T^{-1}\sum_t \|\tau_t-\bar{\tau}\|^2$ relies on $\sum_t(\tau_t-\bar{\tau})=0$, which depends only on $\bar{\tau}$ being the arithmetic mean. The right generalization to H_t-weighted distortion uses $\bar{\tau}_H$ of (4) as the centroid.

Lemma 1 (H_t-weighted max $\geq$ avg identity)**.** *For every admissible* $(\boldsymbol{\tau}, \boldsymbol{H})$ *and every* $w \in \mathbb{R}^d$*,*

$$\tfrac{1}{T}\sum_{t=1}^{T}(w-\tau_t)^\top H_t(w-\tau_t) \;=\; (w-\bar{\tau}_H)^\top \bar{H}(w-\bar{\tau}_H) + \tfrac{1}{T}\sum_{t=1}^{T}(\tau_t-\bar{\tau}_H)^\top H_t(\tau_t-\bar{\tau}_H), \tag{6}$$

and hence $\max_t(w-\tau_t)^\top H_t(w-\tau_t) \geq (w-\bar{\tau}_H)^\top \bar{H}(w-\bar{\tau}_H) + T^{-1}\sum_t(\tau_t-\bar{\tau}_H)^\top H_t(\tau_t-\bar{\tau}_H)$.

The full proof is given in the supplementary material. The argument is a two-line algebraic expansion: the cross-term $\sum_t H_t(\tau_t-\bar{\tau}_H) = T(\bar{H}\bar{\tau}_H - \bar{H}\bar{\tau}_H) = 0$ vanishes by the defining property $\bar{H}\bar{\tau}_H = T^{-1}\sum_t H_t\tau_t$ of the centroid. No probabilistic argument is needed; at $H_t = I_d$ we recover the Phase 0 identity exactly.

Two consequences of Lemma 1 drive everything that follows. First, *components of* w *orthogonal to* $\mathrm{range}(\bar{H}) = V_1 + \cdots + V_T$ *annihilate against every* H_t *and do not affect* (6); without loss of generality the decoder's output lies in an at-most-d_{eff}-dimensional subspace. Second, the max-distortion lower bound separates into a compression term in $\bar{H}$-metric on $\bar{\tau}_H$ and an irreducible floor term that does not depend on w at all. The rest of the argument bounds each piece.

4.2 Closed-form floor

Under the hard distribution $P^\star$ of §3, the floor term admits an exact closed form depending only on d_{eff}.

Lemma 2 (Closed-form floor)**.** *Under* $P^\star$ *with any fixed positive-definite spectra* $(D_t)_{t=1}^T$ *(possibly distinct across tasks),*

$$\mathbb{E}_{P^\star}\left[\tfrac{1}{T}\sum_{t=1}^{T}(\tau_t-\bar{\tau}_H)^\top H_t(\tau_t-\bar{\tau}_H)\right] \;=\; B^2\left(1-\frac{\mathbb{E}[d_{\mathrm{eff}}]}{Tr}\right). \tag{7}$$

The proof (supplementary material) splits into a deterministic identity reducing the floor to a bilinear form in $\bar{\tau}_H$ and an expectation-via-isotropy step. The key probabilistic input is the second-moment identity $\mathbb{E}[H_t \tau_t \tau_t^\top H_t \mid U_t] = (B^2/r) H_t$, which holds *for any* D_t because of the $D_t^{-1/2}$ stretch in the definition of $P^\star$.

Three limiting cases illuminate the formula.

1. **Full-rank limit** ($r = d$, $V_t = \mathbb{R}^d$). Then $d_{\text{eff}} = d$ a.s., floor $= B^2(1 - d/(Td)) = B^2(1-1/T)$, recovering the Phase 0 formula for classical averaged-vector quantization.
2. **Single-task limit** ($T = 1$). Then $d_{\text{eff}} = r$, floor $= B^2(1 - r/r) = 0$, and the rate-distortion function reduces to single-source quantization of $\tau_1 \in V_1 \cong \mathbb{R}^r$. This case matches TurboQuant's Theorem 3 [23].
3. **Shared-subspace limit** ($V_t = V$ for all t). Then $d_{\text{eff}} = r$, floor $= B^2(1 - r/(Tr)) = B^2(1-1/T)$, again recovering Phase 0 with the ambient dimension replaced by the rank.

In the generic LoRA regime (V_t Stiefel-random and $Tr \le d$), $d_{\text{eff}} = Tr$ almost surely and the floor vanishes. The rate-distortion function in this regime is controlled entirely by the compression term.

4.3 Main lower bound

Combining Lemma 1 and Lemma 2 with a rank-r analog of the Phase 0 rotational-invariance argument gives the main theorem.

Theorem 3 (Rank-r RD lower bound, general H_t)**.** *Under $P^\star$ with any fixed positive-definite spectra $(D_t)_{t=1}^T$, and with* $\operatorname{rank}(H_t) = r$ *for all t,*

$$\mathcal{D}^\star(T, d, r, B, R) \;\ge\; B^2\left(1 - \frac{\mathbb{E}[d_{\text{eff}}]}{Tr}\right) \;+\; c_{\text{TQ}} \cdot \frac{B^2 \mathbb{E}[d_{\text{eff}}]}{Tr} \cdot 2^{-2R/\mathbb{E}[d_{\text{eff}}]}. \tag{8}$$

Under Stiefel-random sampling with $Tr \le d$, $d_{\text{eff}} = Tr$ a.s., the floor vanishes, and

$$\mathcal{D}^\star(T, d, r, B, R) \;\ge\; c_{\text{TQ}} \cdot B^2 \cdot 2^{-2R/(Tr)}, \tag{9}$$

with c_{TQ} the universal constant from TurboQuant [23] Theorem 3.

Proof sketch. The argument is a six-step reduction.

Step 1 (Yao's minimax). $\mathcal{D}^\star \ge \inf_{E,D} \mathbb{E}_{P^\star}[\max_t D_t(w^\star)]$ by (5); all subsequent steps bound the RHS.

Step 2 (max $\ge$ avg reduction). By Lemma 1 and Lemma 2,

$$\mathbb{E}_{P^\star}[\max_t D_t(w^\star)] \;\ge\; \mathbb{E}_{P^\star}\left[(w^\star - \bar{\tau}_H)^\top \bar{H}(w^\star - \bar{\tau}_H)\right] + B^2(1 - \mathbb{E}[d_{\text{eff}}]/(Tr)).$$

Step 3 (restrict to $\operatorname{range}(\bar{H})$*).* By the orthogonality remark following Lemma 1, the decoder may be assumed to emit $w^\star \in \operatorname{range}(\bar{H})$ without loss.

Step 4 (change of variable). Let $\eta := \bar{H}^{1/2}\bar{\tau}_H$, viewed as an element of $\mathrm{range}(\bar{H}) \cong \mathbb{R}^{d_{\mathrm{eff}}}$. Then $(w^\star - \bar{\tau}_H)^\top \bar{H}(w^\star - \bar{\tau}_H) = \|\bar{H}^{1/2}w^\star - \eta\|^2_{\mathbb{R}^{d_{\mathrm{eff}}}}$, so the compression subproblem is rate-R quantization of η under squared Euclidean loss in $\mathbb{R}^{d_{\mathrm{eff}}}$.

Step 5 ($O(W)$-equivariance and Shannon bound). Conditioning on $W := \mathrm{range}(\bar{H})$, the joint law of $(U_t, \tau_t)_{t=1}^T$ is $O(W)$-equivariant: for any $\tilde{O} \in O(W)$ extended to $O(d)$ by the identity on $W^\perp$, $\tau_t \mapsto O\tau_t$ and $H_t \mapsto OH_tO^\top$, leaving the marginal joint law unchanged. The pushforward $\eta \mapsto O\eta$ is therefore $O(W)$-invariant in the conditional law. A rotationally-invariant rate-R quantization bound — a rank-r analog of TurboQuant Theorem 3 [23] — gives

$$\mathbb{E}\big[\|\bar{H}^{1/2}w^\star - \eta\|^2 \;\big|\; W\big] \;\geq\; c_{\mathrm{TQ}} \cdot \mathbb{E}\big[\|\eta\|^2 \;\big|\; W\big] \cdot 2^{-2R/d_{\mathrm{eff}}}.$$

The key D_t-freeness of the argument survives in this step because $v_t = BD_t^{-1/2}u_t$ lives in $\mathbb{R}^r$ and is $O(d)$-free regardless of which D_t it uses.

Step 6 (combine). Taking expectation over W and using $\mathbb{E}[\|\eta\|^2] = B^2\mathbb{E}[d_{\mathrm{eff}}]/(Tr)$ (from Lemma 2), substitution into Step 2 yields (8). The Stiefel specialization $Tr \leq d$ gives $d_{\mathrm{eff}} = Tr$ a.s., hence (9).

The full proof is given in the supplementary material; the six-step skeleton above is the complete argument modulo the algebraic expansions in Steps 2 and 6 and the Shannon-bound machinery of Step 5.

Why the general-H_t statement is the headline. Real LoRA fine-tunes have task-specific Hessians with non-trivial spectra. The Fisher information at a fine-tuned point τ_t is rank-r PSD with range in V_t but generally not a projector; its spectrum inside V_t is the task-t curvature and generally differs across tasks. Theorem 3 applies directly to this setting without any modeling assumption on the D_t's beyond positive-definiteness. This closes the MSE $\to$ CE bridge for the lower-bound half of the rate-distortion characterization; the achievability analysis in §5 proceeds under the same general hypothesis.

5 Achievability

The lower bound of §4 is matched by an explicit algorithm in the generic regime (V_t Stiefel-random, $Tr \leq d$, so $d_{\mathrm{eff}} = Tr$ a.s. and the floor in Lemma 2 vanishes). The algorithm is a direct extension of the single-source construction in TurboQuant [23], composed with a centroid-reduction step that uses Lemma 1.

5.1 Algorithm

Encoder and decoder share a pseudorandom seed indexing an orthogonal rotation $O \in O(d_{\mathrm{eff}})$ drawn uniformly from Haar measure (we implement this in practice by the QR decomposition of a $d_{\mathrm{eff}} \times d_{\mathrm{eff}}$ matrix of iid Gaussians).

Encoder.

1. Compute $\bar{H}$, $\bar{\tau}_H$, and an orthonormal basis $Q \in \mathbb{R}^{d \times d_{\text{eff}}}$ of range($\bar{H}$) with eigenvalues $w_1, \ldots, w_{d\text{eff}} > 0$. Form $\eta := \bar{H}^{1/2}\bar{\tau}_H$, represented in the Q-basis as the d_{eff}-vector $\eta_Q = (\sqrt{w_i}\,\langle Q_i, \bar{\tau}_H\rangle)_{i=1\text{eff}}^{d}$.
2. Apply the shared rotation: $\tilde{\eta} := O\eta_Q \in \mathbb{R}^d_{\text{eff}}$.
3. For each coordinate $i \in [d_{\text{eff}}]$, uniform-scalar-quantize $\tilde{\eta}_i$ on $[-c\sigma_{\text{pc}}, c\sigma_{\text{pc}}]$ with b bits, where $\sigma_{\text{pc}} := B/\sqrt{Tr}$ is the per-coordinate standard deviation of $\tilde{\eta}$ and c is a clipping constant (we use $c = 5$). Total rate is $R = bd_{\text{eff}}$ bits.
4. Transmit the quantization indices.

Decoder. Inverts each step: unquantize to $\tilde{\eta}_q$, apply $O^\top$ to get $\eta_{Q,q}$, and emit $w^\star := Q\,\text{diag}(1/\sqrt{w_i})\,\eta_{Q,q} = \bar{H}^{+1/2}\eta_q$ (understood on range($\bar{H}$); components orthogonal to range($\bar{H}$) contribute zero to any H_t-weighted distortion, by the remark following Lemma 1).

The algorithm has three structural features worth naming. First, the centroid $\bar{\tau}_H$ is the *sufficient statistic* for the reconstruction problem in the floor-zero regime — we show below that perfect knowledge of $\bar{\tau}_H$ achieves zero worst-task distortion, so all rate goes to quantizing $\bar{\tau}_H$. Second, the $\bar{H}^{1/2}$ scaling whitens $\bar{\tau}_H$: in the $\eta = \bar{H}^{1/2}\bar{\tau}_H$ coordinates, the per-coordinate second moment is uniform ($B^2/(Tr)$), making scalar quantization on a uniform grid optimal up to constants. Third, the random rotation O is a discrete analog of TurboQuant's Haar-random pre-mixing: it breaks per-coordinate correlations in η_Q and ensures every direction of quantization error contributes equally to $\mathbb{E}\|\delta\eta\|^2$.

5.2 Matching upper bound

Theorem 4 (Matching achievability, floor-zero Stiefel)**.** *Under $P^\star$ with Stiefel-random V_t and $Tr \le d$, for any fixed positive-definite $(D_t)_{t=1}^T$, the algorithm of §5.1 at rate R bits achieves*

$$\mathbb{E}_{P^\star}\big[\max_t (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)\big] \;\le\; C \cdot B^2 \cdot 2^{-2R/(Tr)}, \tag{10}$$

with universal constant $C = Tc^2/3$. Combined with Theorem 3,

$$\mathbb{E}_{P^\star}\big[\max_t (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)\big] \;=\; \Theta\big(B^2 \cdot 2^{-2R/(Tr)}\big),$$

i.e. the Bayesian rate-distortion function under $P^\star$ is pinned down to a universal constant in this regime.

Proof sketch. Five steps.

Step 1 (zero-floor identity). Under $P^\star$ with $Tr \le d$, Lemma 2 gives $\mathbb{E}_{P^\star}[T^{-1}\sum_t (\tau_t - \bar{\tau}_H)^\top H_t(\tau_t - \bar{\tau}_H)] = 0$. Each summand is nonnegative, so almost surely $(\tau_t - \bar{\tau}_H)^\top H_t(\tau_t - \bar{\tau}_H) = 0$ for every t; equivalently, $H_t\bar{\tau}_H = H_t\tau_t$. In this regime $\bar{\tau}_H$ captures every task exactly at infinite rate.

Step 2 (distortion reduces to quantization error on η). Let $\delta w := w^\star - \bar{\tau}_H \in \mathrm{range}(\bar{H})$. Expanding,

$$(w^\star - \tau_t)^\top H_t (w^\star - \tau_t) \;=\; (\delta w)^\top H_t (\delta w),$$

where the constant and cross terms vanish by Step 1 ($H_t(\bar{\tau}_H - \tau_t) = 0$ a.s.). So worst-task distortion is entirely driven by quantization error on $\bar{\tau}_H$ (equivalently on η, after whitening).

*Step 3 (*max $\le T \cdot$ avg*).* Summing and using $\sum_t H_t = T\bar{H}$,

$$\max_t (\delta w)^\top H_t (\delta w) \;\le\; \sum_t (\delta w)^\top H_t (\delta w) \;=\; T(\delta w)^\top \bar{H} (\delta w) \;=\; T\|\bar{H}^{1/2}\delta w\|^2 \;=\; T\|\delta\eta\|^2,$$

where $\delta\eta := \eta_q - \eta$ is the quantization error in η-space. This is the achievability-side analog of Lemma 1: max-distortion is controlled by the $\bar{H}$-weighted average, which is itself $\|\delta\eta\|^2$.

Step 4 (scalar-quantization error under the random mixer). After the rotation O, each coordinate $\tilde{\eta}_i$ has second moment $\mathbb{E}[\tilde{\eta}_i^2] = \mathbb{E}\|\eta\|^2/d_{\mathrm{eff}} = B^2/(Tr)$ (by Lemma 2 and the isotropy induced by O). Uniform scalar quantization on a grid of width $\Delta = 2c\sigma_{\mathrm{pc}}/2^b$ has per-coordinate MSE at most $\Delta^2/12 = c^2\sigma_{\mathrm{pc}}^2 2^{-2b}/3$, giving

$$\mathbb{E}\|\delta\tilde{\eta}\|^2 \;\le\; d_{\mathrm{eff}} \cdot \frac{c^2 B^2}{3Tr} \cdot 2^{-2b} \;=\; \frac{c^2 B^2 d_{\mathrm{eff}}}{3Tr} \cdot 2^{-2R/d_{\mathrm{eff}}}.$$

Orthogonality of O preserves $\|\cdot\|^2$, so $\|\delta\eta\|^2 = \|\delta\tilde{\eta}\|^2$. In the Stiefel regime $d_{\mathrm{eff}} = Tr$, yielding $\mathbb{E}\|\delta\eta\|^2 \le (c^2B^2/3) \cdot 2^{-2R/(Tr)}$.

Step 5 (combine). Substituting Step 4 into Step 3,

$$\mathbb{E}_{P^\star}[\max_t (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)] \;\le\; T\,\mathbb{E}\|\delta\eta\|^2 \;\le\; \frac{Tc^2}{3} \cdot B^2 \cdot 2^{-2R/(Tr)}.$$

This is (10) with $C = Tc^2/3$. □

Empirical calibration. Monte Carlo experiments report $\mathbb{E}_{P^\star}[\max_t D_t(w^\star)]$ across $T \in \{2,3,4\}$, $r \in \{4,8,16\}$, $d \in \{128,512,2048\}$, and both isotropic and heavily-anisotropic (D_t) spectra. Empirically, the ratio of measured distortion to the Theorem 3 lower bound at $c_{\mathrm{TQ}} = 1$ is ≤ 13 across all tested configurations and rate points, constant within bootstrap CI. This confirms the $\Theta(2^{-2R/(Tr)})$ rate and that the upper-bound constant $C = Tc^2/3$ is tight up to the choice of clipping c. See §6 for details.

5.3 The shared-V / floor-positive regime

Theorem 4 covers the generic regime. When the V_t's overlap substantially (so that $d_{\mathrm{eff}} < Tr$ and the floor is strictly positive), the algorithm above still applies but the proof argument of Step 1 (zero-floor identity) fails. In the floor-positive regime we construct a refinement that quantizes around the Chebyshev

center of the ensemble in an active-set-aware null-space split, achieving excess distortion of order $2^{-2R/(d_{\text{eff}}+|A|-1)}$, where $|A|$ is the size of the active set at the Chebyshev center. This is strictly better than the naive $2^{-R/d_{\text{eff}}}$ rate of a direct scalar quantization but strictly worse than the lower-bound exponent $2^{-2R/d_{\text{eff}}}$ of Theorem 3; the remaining exponent gap is discussed in Remark 5. We have numerically validated the construction across $T \in \{2, 3, 4\}$ and the full range of anisotropy regimes (iso+iso, iso+geom, geom+twoscale, lin+twoscale, geom+iso); see §6.

Closing the shared-V exponent gap appears to require non-linear / random-coding encoders: a random-codebook encoder achieves slopes strictly steeper than our explicit linear construction in the shared-V regime, so the true rate-distortion function in that regime lies strictly between the two bounds. A rigorous sharpening of either the lower or upper bound in this regime is left open.

Remark 5 (Shared-V exponent gap). In the floor-positive regime, the explicit linear null-space split of §5.3 achieves excess distortion $\Theta(2^{-2R/(d_{\text{eff}}+|A|-1)})$. Valid rate-$R$ random-codebook encoders empirically achieve strictly steeper slopes than this linear construction (we verify this for $T = 3, r = 3$, where the random-codebook slope is -1.45 versus the linear prediction -1.20). The lower bound of Theorem 3 gives $\Omega(2^{-2R/d_{\text{eff}}})$, which is also not attained by our construction when $|A| > 1$. Neither bound is tight; the exact rate-distortion exponent in this regime is open.

6 Experiments

We report Monte Carlo experiments on rank-r LoRA-like ensembles validating the theoretical rates and constants. All runs use the hard distribution $P^\star$ of §3 with task-dependent spectra $(D_t)_{t=1}^T$. Code, random seeds, and raw outputs are released with the paper.

6.1 Synthetic validation

Lower-bound slope. We sample ensembles across $T \in \{2, 3, 4, 8\}$, $r \in \{4, 8, 16\}$, $d \in \{128, 512, 2048\}$, with both isotropic spectra ($D_t = I_r$) and four heavy-anisotropy families (geometric, linear, two-scale, and mixed; spectra spread across $[2^{-1}, 2^1]$). For each configuration we measure the excess $\mathbb{E}_{P^\star}[\|\bar{H}^{1/2}w^\star - \eta\|^2]$ of the Gaussian-QR + uniform scalar quantization algorithm of §5 as a function of $R = bd_{\text{eff}}$ bits, and fit a linear regression of $\log_2 \mathbb{E}[\text{excess}]$ against $b = R/d_{\text{eff}}$. Across all tested configurations the empirical slope is -2.00 ± 0.01 (95% bootstrap CI across 1000 trials per configuration), matching the prediction $-2R/d_{\text{eff}}$ of Theorem 3. The floor term $B^2(1 - \mathbb{E}[d_{\text{eff}}]/(Tr))$ is reproduced to four to five digits of precision in the common-D_0 and task-dependent-D_t regimes.

Achievability constant. We measure the ratio $\mathbb{E}_{P^\star}[\max_t D_t(w^\star)]/(c_{\text{TQ}} B^2 2^{-2R/(Tr)})$ of empirical upper bound to the Stiefel lower bound of Theorem 3 at $c_{\text{TQ}} = 1$.

The ratio is constant in R — a necessary condition for the UB to match the LB in order — and lies in $[11, 13]$ across $T \in \{2, 3\}$, $r = 4$, $d = 128$, for isotropic and task-mismatched anisotropic spectra. This is consistent with the analytical constant $C = Tc^2/3 \approx 17$ at $T = 2$, $c = 5$, combined with $c_{\mathrm{TQ}} \approx 1$ from TurboQuant [23].

Shared-V regime. For the floor-positive regime we implement the null-space-aware bit allocation of §5.3 with per-coord fractional bits and a locked clip $c = 11.5\sigma_{\mathrm{pc}}$ (tuned once, applied across all configurations). At $T = 2$, $r = 4$, across five anisotropy mixes (iso+iso, iso+geom, geom+twoscale, lin+twoscale, geom+iso), the empirical slope of excess over cheb^2 is -1.60 ± 0.10 at $n_{\mathrm{trials}} = 1000$, matching the theoretical prediction $-2r/(r + |A| - 1) = -8/5$ with bootstrap 95% CI ± 0.010 per cell. The ± 0.10 spread reflects structural anisotropy: iso+iso tracks closer to the scalar-quantization limit -2; geom+twoscale undershoots due to H_t-metric distortion on the perpendicular subspace. Extending to $T = 3, 4$ via a SOCP Chebyshev solver with Gauss-Newton KKT refinement (residual $\sim 10^{-15}$), the null-space split continues to track the theoretical prediction across mixed anisotropies.

Random-codebook comparison. As a sanity check on the tightness of our linear construction in the shared-V regime, we ran a valid rate-R random-codebook encoder (2^R iid Gaussian codewords, encoder picks the one minimizing $\max_t D_t$, $R \leq 18$). At $T = 3$, $r = 3$, the random-codebook slope is -1.45, strictly steeper than our explicit linear construction's -1.20. This rules out sharpening the lower bound to match the linear upper-bound exponent; the true rate-distortion function in the floor-positive regime lies strictly between $-2r/r = -2$ and $-2r/(r + |A| - 1)$ and is left open (Remark 5).

6.2 Real-LLM merging

A calibration of Theorem 3 against real LoRA fine-tunes (measuring worst-task distortion on merged Qwen / Gemma / Phi adapters for standard merging methods such as Task Arithmetic, TIES, DARE, KnOTS, and TVQ) is deferred to a future version of this paper. The aim of such experiments is to test whether the bound predicts the *ranking* of merging methods across tasks of varying subspace overlap, not merely to reproduce theoretical slopes on idealized data. We view this as a substantive empirical contribution in its own right and separate it from the present theoretical treatment.

7 Discussion and Limitations

Cross-entropy extension. Theorem 3 is proved in the quadratic-loss setting with H_t a positive-semidefinite rank-r matrix. For LLM cross-entropy loss, the natural H_t is the Fisher information matrix, which is rank-r PSD near a LoRA fine-tuned point but not a projector. Our proof uses only the per-task

identity $H_t H_t^+ H_t = H_t$ and the $O(d)$-equivariance of the hard distribution, both of which survive verbatim under the Fisher-metric interpretation (the D_t-freeness of Step 5 is the key). A fully rigorous MSE-to-CE bridge via local quadratic approximation requires controlling the second-order remainder across the H_t-ball; we sketch this extension in the supplementary material and leave a quantitative version for future work.

The shared-V exponent gap. In the floor-positive regime (e.g. shared or substantially-overlapping V_t), the explicit linear construction of §5 achieves an excess distortion decay of $2^{-2R/(d_{\text{eff}}+|A|-1)}$, slower than the lower bound's $2^{-2R/d_{\text{eff}}}$. We verified numerically that a random-codebook (non-linear) encoder achieves slopes strictly steeper than the linear construction, so the truth lies *strictly between* the two bounds. Closing this gap — either by sharpening the lower bound or by constructing a non-linear encoder that matches the $2^{-2R/d_{\text{eff}}}$ rate — is a concrete open problem that we view as beyond the scope of this work. Remark 5 elaborates.

Scope of the hard distribution. The lower bound is established against $P^\star$ (Stiefel-random V_t, arbitrary task-dependent spectra). Real LoRA fine-tunes exhibit structure not captured by iid Stiefel random sampling: tasks that share semantic similarity produce correlated subspaces; fine-tuning induces anisotropic curvature; and the full-dimensional backbone of the base model introduces rank-d modes the bound ignores. Extending the lower bound to structured task distributions, or supplying a second construction that exploits task correlation to beat our generic upper bound, are natural directions. Until such extensions exist, the theorems of this paper should be read as characterizing the *information-theoretic* floor for merging, not a tight prediction of any specific real-LLM merging experiment.

Practical implications. Two guidance points for practitioners follow from the theorems. (i) The floor $B^2(1 - d_{\text{eff}}/(Tr))$ identifies when rate is wasted: if tasks share substantial subspace overlap, no merging algorithm closes the floor and additional bits buy little. This gives an operational criterion for deciding when to merge at all versus when to serve tasks from separate adapters. (ii) The compression term $B^2(d_{\text{eff}}/(Tr))2^{-2R/d_{\text{eff}}}$ is matched by an explicit algorithm that runs in $O(d_{\text{eff}}^2)$ time per merge (the Gaussian-QR rotation dominates). For LoRA with $d_{\text{eff}} = Tr$ in the thousands, this is negligible compared with the merging overhead of existing heuristics; deployment cost is not a blocker.

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